

8.2 ECOSYSTEM- ELEMENTARY ASPECTS

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An 'ecosystem' consists of both **structural** components and **functional** components. The structural components of ecosystem are of two types – **abiotic** and **biotic** factors. Abiotic factors are again of two types- physical and chemical factors. The physical factors are light, temperature, soil, pressure, etc., Oxygen, carbon dioxide and the various mineral nutrients in the soil / water constitute the chemical factors. These physical and chemical factors affect the life of the organisms present in an ecosystem. Energy transfer and recycling of minerals constitute the functional aspect of an ecosystem.

8.2.1 Light

Since plants produce food through photosynthesis, a process which is possible only when sunlight is available as a source of energy, we can easily understand the importance of light for all living organisms, particularly the 'autotrophs'. Many species of small plants (herbs and shrubs) growing in forests are adapted to photosynthesize optimally under very low light conditions because they are constantly overshadowed by tall trees with well spread branches. Many plants are also dependent on sunlight to meet their **photoperiod requirement** for flowering.

For many animals, light is important in that they use the diurnal and seasonal variations in the **intensity** and **duration (photoperiod)** of light, as cues (indicators) for timing their foraging (collection of food), reproductive and migratory activities. The availability of light on land is closely linked with that of temperature since the sun is the source of both. But, deep in the oceans, the environment is always dark and its inhabitants do not experience light. The spectral quality of solar radiation is also important for life. The UV component of the spectrum is harmful to many organisms. All the colour components of the visible spectrum are not available for marine plants, living at different depths of the seas.

I. Biological effects of light

Light is an ecological factor that shows its influence on many biological phenomena and activities like pigmentation, movement, vision, metabolism, etc.,

1. Effect of light on pigmentation

Light influences the colour of the skin. The animals which live in regions of low intensity of light such as caves, have less pigmentation than that of the animals exposed to light. For example a cave dwelling amphibian, **Proteus anguinus**, has pale coloured skin because of less pigmentation. When it is brought into sunlight the skin colour turns *dark* over a period of time.

2. Effect of light on direction and rate of movement

Response shown by an organism to changes in light is called **photoresponse**. The major photoresponses of motile organisms include:

- i) **Phototaxis** is oriented locomotor movement of an organism towards or away from the direction of light as seen in euglenas(positive response), cockroaches(negative response) and,
- ii) **Photokinesis** is the influence of light on non- directional movement of organisms as seen in the larvae of *Pinnotheres macculatus*- the mussel crab, in which intensity of light influences the velocity of the movement of the organism.

Phototropism: The orientation to light of a non-motile organism such as a plant or plant part is called phototropism. For example the growth mediated bending of plants towards sunlight is due to phototropism.

3. Influence of Light on Vision and Behavior

Light is essential for vision, to procure food, self protection, identification and selection of the mate for sexual reproduction. It also influences the behaviour of organisms. Some animals are active during the 'day time' (**diurnal** animals e.g. majority of the birds, reptiles and mammals. Others, which are active during night time are called **nocturnal** animals e.g. earthworms, cockroaches, etc.,

4. Effect of light on metabolism

The intensity of light influences the rate of metabolism in animals. At lower intensities of light the rate of metabolism slows down, where as in higher intensities it increases.

5. Effect of light on biological rhythms

By definition rhythm is a change that is repeated with similar pattern in a time frame. In the bodies of organisms many behavioural activities are repeated at regular intervals and these are called 'biological rhythms'.

Biological rhythms that occur in a time period of 24 hours are **circadian rhythms**. If the biological rhythms repeat annually (every year), they are called **circannular rhythms**. They are controlled by '**biological clocks**' present in many living organisms. Light has a role in setting or resetting the biological clocks.

6. Effect of 'photoperiod' on animals (photoperiodism)

The duration of the light hours / exposure to light in a day is known as **photoperiod**. The response of organisms for the photoperiod is called **photoperiodism**. The response to photoperiod often centers on reproduction and survival(production of flowers, the migration of birds etc.,). The specific day length which is essential for the

initiation of seasonal events is called '**critical photoperiod**'. The length of critical photoperiod is not the same in all the species, it varies from species to species and also within the same species inhabiting at various latitudes. For example, when winter sets in Siberia, the light available time period in a day decreases, and so some birds migrate all the way to the new **feeding** and **breeding grounds** in various parts of India. When it is summer time back in Siberia, they migrate back to their homeland.

Exposure of young fowls in poultry to more lighted hours in a day induces speedy maturity and the egg laying process. It is a familiar example to most of us.

Bioluminescence: Production of light by certain living organisms is called bioluminescence. The light emitted by living organisms is devoid of infrared rays, and so it is called '*cold light*'. Some Jelly fishes (cnidarian), *Chaetopterus* (annelid), firefly (arthropod), squids (mollusc), *Pyrosoma* (protochordate), and some fishes produce bioluminescence. It is believed to *influence life* in the dark habitat by providing intra-specific communication, sexual attraction, attracting the prey, sending protective warning signals, etc.

II. Effect of UV Rays

The short wave lengths of light ranging from 100 nm to 380nm are called **Ultraviolet rays**. UV radiation is classified into three categories: UVC light (100nm to 280 nm), UVB (280nm to 320nm) and UVA (320nm to 380nm). UV radiation kills the microorganisms present on the body surfaces of animals. UV radiation helps in the conversion of sterols present in the skin into vitamin D in mammals. Compared to UVA rays, UVB and UVC rays are more harmful to organisms.

8.2.2 Temperature

Temperature is a measure of the intensity of heat. The temperature on land or in water is not uniform. On land the temperature variations are more pronounced when compared to the aquatic medium, because land absorbs or loses heat much quicker than water. The temperature on land depends on seasons and the geographical area on this planet. Temperature decreases progressively when we move from the equator to the poles. Altitude also causes variations in temperature. For instance, the temperature decreases gradually as we move to the top of the mountains.

I. Thermal Stratification

Temperature variations occur with seasonal changes in the temperate regions. These differences in the temperature form 'thermal layers' in water. These phenomena are called **thermal stratifications**.



Summer stratification

During summer in temperate lakes, the density of water decreases with the increase in its temperature (21-25°C). This 'upper layer' is called the epilimnion. Below the epilimnion there is a zone in which the temperature decreases at the rate of 1°C per meter in depth, and it is called the thermocline. The bottom layer is the **hypolimnion**, where water is cold and has low oxygen content (*due to absence of photosynthetic activity*).

During **autumn** (also called **Fall**), the epilimnion cools down, and the surface water becomes heavy when the temperature is 4°C, and sinks to the bottom of the lake. Overturns bring about 'uniform temperature' in lakes during that period. This circulation during the autumn is known as the **fall overturn** or **autumn overturn**. The upper oxygen rich water moves to the bottom, and the bottom water moves to the surface.



Winter stratification /stagnation

The diagram illustrates the seasonal cycle of lake mixing through four panels arranged in a square, connected by a central circular arrow indicating a continuous cycle.

- Winter:** Shows a cross-section of a lake with a single layer of cold water. The water is dark blue, indicating low temperature. The shoreline is brown with some sparse green vegetation.
- Spring:** Shows the lake beginning to mix. A circular arrow indicates the movement of water. The water is a lighter blue, suggesting warming and mixing.
- Summer:** Shows the lake in a state of strong stratification. The water is divided into three distinct layers:
 - Epilimnion:** The top layer of warm, light blue water.
 - Thermocline:** The middle layer where temperature changes rapidly with depth.
 - Hypolimnion:** The bottom layer of cold, dark blue water.
- Autumn:** Shows the lake beginning to mix again. A circular arrow indicates the movement of water. The water is a medium blue, suggesting warming and mixing.

A central circular arrow with a cross in the middle connects the four panels, indicating the continuous cycle of seasonal mixing.

cool (4°C) water occupies the lake. The aquatic animals continue their life below the icy layer. At lower temperatures the activity of bacteria and the rate of oxygen consumption by aquatic animals decrease. Hence, organisms can survive below the frozen upper water without being subjected to ‘**hypoxia**’ (low oxygen availability). In the ‘Spring season’ the temperatures start rising. When it reaches 4°C , the

Fig. 8.6 Stratification in temperate lakes

water becomes more dense and heavy and sinks to the bottom, taking 'oxygen rich water' to the bottom. The upper oxygen rich water sinks down and the bottom 'nutrient rich water' reaches the surface. It is called '**spring overturn**'. The lakes which show overturns twice a year are called '**dimictic lakes**'. Thus 'stratifications' and 'overturns' help survival of organisms at all levels in deep lakes, as they help in redistribution of oxygen.

II. Biological effects of Temperature

1. Temperature Tolerance

A few organisms can tolerate and thrive in a wide range of temperatures they are called **eurythermal** (e.g. all homoeotherms), but, a vast majority of organisms are restricted to a narrow range of temperatures (such organisms are called **stenothermal** e.g. fishes and coral animals). The levels of thermal tolerance of different species determine their geographical distribution.

2. Temperature and Metabolism

Temperature affects the *working of enzymes* and through it, the **basal metabolism**, and other physiological functions of organism. The temperature at which the metabolic activities occur at the climax level is called the '**optimum temperature**'. The lowest temperature at which an organism can live indefinitely is called **minimum effective temperature**. If an animal or plant is subjected to a temperature below the minimum effective limit, it enters into a condition of inactiveness called **chill coma**. The metabolic rate increases with the rise of temperature from the **minimum effective temperature** to **optimum temperature**.

The maximum temperature at which a species can live indefinitely in an active state is called **maximum effective temperature**. If the temperature is raised above the maximum effective temperature, the animals enter into '**heat coma**'. The maximum temperature varies much in different animals.

3. van't Hoff's rule

van't Hoff, a Nobel Laureate in thermochemistry, proposed that, with the increase of every 10°C , the rate of metabolic activities doubles. This rule is referred to as the **van't Hoff's rule**. van't Hoff's rule can also be stated in reverse saying that the reaction rate is halved with the decrease of every 10°C . The effect of temperature on the rate of a reaction is expressed in terms of **temperature coefficient** or Q_{10} value. Q_{10} values are estimated taking the ratio between **the rate of a reaction at $X^{\circ}\text{C}$** and **rate of reaction at $(X-10^{\circ}\text{C})$** . In the 'living systems' the Q_{10} value is about 2.0. If the Q_{10} value is 2.0, it means, for every 10°C increase, the rate of metabolism doubles.

4. Cyclomorphosis

The cyclic seasonal morphological variations among certain organisms is called **cyclomorphosis**. This phenomenon has been demonstrated in the cladoceran (a sub group of crustacea) *Daphnia* (*water flea*). In the winter season the head of *Daphnia* is

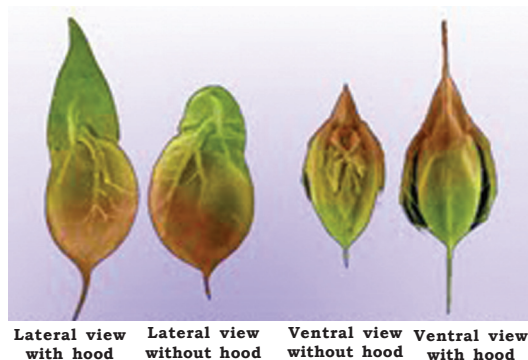


Fig. 8.7 Cyclomorphosis in *Daphnia*

‘round’ in shape (typical or non-helmet morph). With the onset of the spring season, a small ‘helmet’/‘hood’ starts developing on it. The helmet attains the maximum size in summer. In ‘autumn’ the helmet starts receding. By the winter season, the head becomes round. Some scientists are of the opinion that **cyclomorphosis** is a seasonal adaptation to changing densities of the water in lakes – in summer as the water is less dense *Daphnia* requires a larger body surface to keep floating easily. During winter the water is more dense, and so it does not require a larger surface area of the body to keep floating. Others

believe that these cyclic changes are adaptations to ‘**stabilize the movement**’ in water. Compared to the ‘typical morphs’, the ‘helmeted morphs’ can **resist the water currents better** to stay in the water rich in food materials.

5. Temperature adaptations

Temperature adaptations in animals can be dealt under three heads a) Behavioural adaptations, b) Morphological and Anatomical adaptations and c) Physiological adaptations.

a. Behavioural adaptations

Some organisms show behavioural responses to cope with variations in their environment. **Desert lizards** manage to keep their body temperature fairly constant by behavioural means. They ‘bask’ (staying in the warmth of sunlight) in the sun and absorb heat when their body temperature drops below the comfort zone, but move into shade when the temperature starts increasing. Some species are capable of burrowing into the soil to escape from the excessive heat above the ground level.

b. Morphological and anatomical adaptations

In the polar seas, aquatic mammals such as the seals have a thick layer of fat (**blubber**) that acts as an **insulator** and reduces the loss of body heat, underneath their skin.

The animals which live in colder regions have larger body size with greater mass. The body mass is useful to generate more heat. As per **Bergmann’s rule** mammals and other warm blooded animals living in colder regions have less ‘**surface area to body volume ratio**’, than their counterparts living in the tropical regions.

The small surface area helps to conserve heat by reducing the surface area of heat loss. For instance, the body size of American moose / Eurasian elk (*Alces alces*), increases with the latitudes in which they live. ^[1]Moose of northern part of Sweden shows 15-20% more **body mass** than the same species (counterparts) living in the southern Sweden.

Mammals from colder climates generally have shorter earlobes and limbs (extremities of the body) to minimize heat loss. Large earlobes and long limbs increase the surface area without changing the body volume. This is known as **Allen's rule**. For instance, the polar fox, *Vulpes lagopus* (formerly called *Alopex lagopus*), has short extremities to minimize the heat loss from the body. In contrast, the desert fox, *Vulpes zerda*, has large ear lobes and limbs to facilitate better 'heat loss' from the body.

c. Physiological adaptations

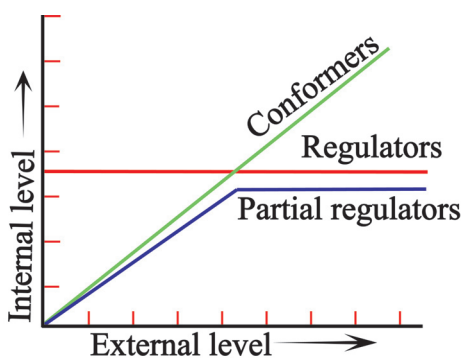
In most animals, all the physiological functions proceed 'optimally' in a narrow temperature range (in humans, it is 37°C). However there are microbes (**archaebacteria**) that flourish in hot springs and in some parts of deep seas, where temperatures far exceed 100°C. Many fish thrive in Antarctic waters where the temperature is always below zero. Having realized that the abiotic conditions of many habitats may vary over a time period, we now ask - **How do the organisms living in such habitats manage with stressful conditions?** One would expect that during the course of millions of years of their existence, many species would have evolved a relatively constant internal (within the body) environment. It permits all biochemical reactions and physiological functions to proceed with maximal efficiency and thus, enhance the overall 'fitness' of the species. This constancy, could be chiefly in terms of **optimal temperature** and **optimal osmotic concentration** of body fluids. So the organism should try to maintain the constancy of its internal environment (homeostasis) despite varying external environmental conditions that tend to upset its homeostasis. This is achieved by the processes described below.

(i) Regulate

Some organisms are able to maintain homeostasis by physiological (sometimes behavioural also) means which ensures constant body temperature, constant osmotic concentration, etc., All birds and mammals, and a very few lower vertebrate and invertebrate species are indeed capable of such regulation (**thermoregulation** and **osmoregulation**). Evolutionary biologists believe that the 'success' of mammals is largely due to their ability to maintain a constant body temperature and thrive whether they live in Antarctica or in the Sahara desert.

The mechanisms used by most mammals to regulate their body temperature are similar to the ones that we, the humans use. We maintain a constant body

[1] The traditional example of comparison between the large sized polar bear and the small sized tropical black bear to support Bergmann's rule, is not acceptable to some Modern Ecologists as these bears belong to two different Genera, altogether (i.e., not the same species to be compared).



temperature of 37°C . In summer, when the outside temperature is more than our body temperature, we sweat profusely. The resulting '**evaporative cooling**' brings down the body temperature. In winter when the temperature is much lower than 37°C , we start to **shiver** (a kind of exercise which produces heat and raises the body temperature – a type of body's own defence mechanism against low temperatures). Plants, on the other hand, do not have such mechanisms to maintain internal temperatures.

(ii) Conform^[2]

Majority (99 percent) of animals cannot maintain a constant internal environment. Their body temperature changes with the ambient (surrounding) temperature. In aquatic animals, the osmotic concentration of the body fluids changes along with that of the surrounding water. Such animals are described as '**conformers**'.

(iii) Partially regulate^[3]

Animals such as 'camels' can be 'conformers' up to a particular range of temperature and 'regulator' afterwards. So, they are described as 'partial regulators' or 'partial conformers'.

Why did not the conformers evolve to become regulators?

Thermoregulation is energetically 'expensive' for many organisms. This is particularly true in small animals like shrews and humming birds. Heat loss or heat gain is a function of the surface area. Since small animals have a larger surface area relative to their volume, they tend to lose body heat very fast when it is cold outside; then they have to spend much energy to generate body heat through metabolism. This is the main reason why very small animals are rarely found in the 'polar regions'. During the course of evolution, the costs and benefits of maintaining a constant internal environment are taken into consideration. Some species have evolved the ability to regulate, but only over a limited range of environmental conditions, beyond which they simply conform.

If the stressful external conditions are localized or remain only for a short duration, the organism has two other alternatives.

(iv) Migrate

The organism can move away temporarily from the 'stressful habitat' to a more 'hospitable' (comfortable) area and return when the stressful period is over. In human

^[2] Adapt from one condition to a new or different conditions.

^[3] There are no perfect regulators or perfect conformers. There are different intermediate grades among animals.

analogy (comparison), this strategy is comparable a person moving from Delhi to Shimla for the duration of summer. Many animals, particularly birds, during winter undertake long-distance migrations to more hospitable areas. Every winter, many places in India including the famous Keoladeo or **Keoladeo Ghana National Park** (Formerly - Bharatpur bird sanctuary) in Rajasthan and Pulicat Lake in Andhra Pradesh host thousands of 'migratory birds' coming from Siberia and other extremely cold northern regions.

(v) Suspend life activities

In bacteria, fungi and lower plants, various kinds of thick-walled spores are formed which help them survive unfavourable conditions. They germinate (come out of the spore wall and produce a normal active organism) on the return of suitable environmental conditions.

Plant Examples : In higher plants, seeds and some other vegetative reproductive structures serve as means **to tide over periods of stress** besides helping in dispersal - they germinate to form new plants under favourable moisture and temperature conditions. They do so by reducing their metabolic activity and going into a state of 'dormancy'.

Some animals can avoid the stress by escaping in 'time' (migration is- escaping in 'space'). The familiar case of 'polar bears' going into **hibernation** during winter is an example of **escape in time**. Some snails, fish and frogs go into **aestivation** to avoid summer-related problems-heat and desiccation.

6. Diapause

Certain organisms show delay in development, during periods of unfavourable environmental conditions and spend some period in a state of 'inactiveness' called '**diapause**'. This dormant period in animals is a mechanism to survive extremes of temperature, drought, etc. It is seen mostly in insects and embryos of some fish. Under unfavourable conditions many zooplankton species in lakes and ponds are known to enter diapause (Ref:NCERT Text Book).

8.2.3 Water

Water is another important factor influencing the life of organisms. Life is unsustainable without water. Its availability is so limited in deserts that only certain **special adaptations** make it possible for them to live there. You might think that organisms living in oceans, lakes and rivers should not face any water-related problems, but it is not true. For aquatic organisms the quality (chemical composition, pH, etc.) of water becomes important. **The salt concentration (measured as salinity in parts per 1000) is less than 5 in inland waters, and 30-35 in the seawater.** Some organisms are tolerant to a wide range of salinities (**euryhaline**), but others are restricted to a narrow range (**stenohaline**). Many freshwater animals cannot live for long in sea water and vice versa because of the osmotic problems, they would face.

1. Adaptations in freshwater habitat

Animals living in freshwaters have to tackle the problem of **endosmosis**. The osmotic pressure of freshwater is very low and that of the body fluids of freshwater organisms is much higher. So water tends to enter into bodies by endosmosis. To maintain the balance of water in the bodies, the freshwater organisms acquired several adaptations such as, contractile vacuoles in the freshwater protozoans, large **glomerular kidneys** in fishes, etc., They send out large quantities of urine, along which some salts are also lost. To compensate the 'salt loss' through urine, freshwater fishes have 'salt absorbing' '**chloride cells**' in their gills. The major problem in freshwater ponds is - in summer most of the ponds dry up. To overcome this problem most of the freshwater protists undergo **encystment**. The freshwater sponges produce asexual reproductive bodies, called **gemmules**, to tide over the unfavourable conditions of the summer. The 'African lungfish', *Protopterus*, burrows into the mud and forms a 'gelatinous cocoon' around it, to survive, in summer.

2. Adaptations in marine habitat

Seawater is high in salt content compared to that of the body fluids. So, the marine animals continuously tend to lose water from their bodies by **exosmosis** and face the problem of dehydration. To overcome the problem of water loss, marine fishes have **aglomerular kidneys** with less number of nephrons. Such kidneys minimize the loss of water through urine. To compensate water loss the marine fish drink more water, and along with this water, salts are added to the body fluids and disturb the internal equilibrium. To maintain salt balance (**salt homeostasis**) in the body, they have **salt secreting chloride cells** in their **gills**. Marine birds like **sea gulls** and **penguins** eliminate salts in the form of salty fluid that drips through their nostrils. In turtles the ducts of chloride secreting glands open near the eyes. Some cartilaginous fishes retain **urea** and **trimethylamine oxide** (TMO) in their blood to keep the body fluid **isotonic** to the sea water and avoid dehydration of the body due to exosmosis.

3. Water related adaptations in brackish water animals

The animals of brackish water are adapted to withstand wide fluctuations in salinity. Such organisms are called **euryhaline** animals. The migratory fishes such as **salmon** and **Hilsa** are **anadromous** fishes i.e., they migrate from the sea to freshwater, for breeding ; *Anguilla bengalensis* is a **catadromous** fish i.e., it migrates from the river to sea, for breeding. In these fishes their **glomerular kidneys** are adjusted to changing salinities. The chloride cells are adapted to **excrete** or **absorb** salts depending on the situation. On entering the river salmon drinks more freshwater to maintain the concentration of body fluids equal to that of the surround water.

4. Water related adaptations for terrestrial life

In the absence of an external source of water, the **kangaroo rat** of the North American deserts is capable of meeting all its water requirements through oxidation of its internal fat (in which water is a by product-*metabolic water*). It also has the ability to concentrate its urine, so that minimal volume of water is lost in the process of removal of their excretory products.

Plant Example: Many desert plants have a thick cuticle on their leaf surfaces and have their stomata arranged in deep pits to minimize water loss through transpiration. They also have a special photosynthetic pathway (⁴CAM) that enables their stomata to remain closed during day time. Some desert plants like *Opuntia*, have no leaves - they are reduced to spines -and the photosynthetic function is taken over by the flattened stems to reduce loss of water (Ref: NCERT Text BookK).

8.2.4 Soil

The nature and properties of soil in different places vary depending on the climate, and the '*weathering*⁵' processes involved. Various characteristics of the soil such as soil composition, grain size and aggregation determine the percolation and water holding capacity of the soils. These characteristics, along with the parameters such as pH, mineral composition etc., determine to a large extent the vegetation in any area. This in turn dictates the type of animals that can be supported. Similarly, in the aquatic environment, the sediment-characteristics often determine the type of benthic animals that can live there.

8.2.5 Pressure

Pressure is another factor that changes dramatically with depth in the ocean. Organisms on land face less than one 'atmosphere' of pressure at the sea level. Since water is much heavier than air, marine organisms are under much more pressure than those on land. The pressure in water increases at the rate of 1 atmosphere per 10m depth. The organisms living in such extreme environments show a wide range of biochemical adaptations. Some organisms possess adaptations that are physiological and allow them to respond quickly to a stressful situation. If you had ever been to any high altitude place (e.g. >3,500m **Rohtang Pass** near **Manali** and **Manasarovar**, in Tibet) you must have experienced what is called **altitude sickness**. Its symptoms include **nausea** (vomiting sense), **fatigue** (tiredness) and **heart palpitations** (abnormality in heart beat). This is because in the low atmospheric pressure of high altitudes, the body does not get enough oxygen. But, you gradually get acclimatized and overcome the altitude sickness. How did your body solve this problem? The body compensates low oxygen availability by increasing red blood cell production and increasing the rate of breathing. (Note: **decreasing the binding capacity of haemoglobin** as mentioned in *NCERT Text Book, Page-226*).

[⁴] Crassulacean Acid Metabolism Pathway.

[⁵] Weathering is a collective term for the processes by which rock at or near the earth's surface is disintegrated and decomposed by the action of wind, water etc., to form soil.